Theorem 1.3

Let A be square matrix of order n. Then, A^{-1} exists if and only if A is non-singular.

Proof

Suppose that A^{-1} exists. Then $AA^{-1} = A^{-1}A = I_n$.

By the product rule for determinants, we get

$\det(AA^{-1}) = \det(A)\det(A^{-1}) = \det(A^{-1})\det(A) = \det(I_n) = 1$. So, $|A| = \det(A) \neq 0$.

Hence A is non-singular.

Conversely, suppose that A is non-singular.

Then $|A| \neq 0$. By Theorem 1.1, we get

$$A(\text{adj } A) = (\text{adj } A)A = |A| I_n.$$

So, dividing by $|A|$, we get $A\left(\frac{1}{|A|}\text{adj } A\right) = \left(\frac{1}{|A|}\text{adj } A\right)A = I_n$.

Thus, we are able to find a matrix $B = \frac{1}{|A|}\text{adj } A$ such that $AB = BA = I_n$.

Hence, the inverse of A exists and it is given by $\boldsymbol{A^{-1} = \frac{1}{|A|}\text{adj } A.}$

Remark

The determinant of a singular matrix is 0 and so a singular matrix has no inverse.

Example 1.2

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is non-singular, find A^{-1}.

Solution

We first find adj A. By definition, we get $\text{adj } A = \begin{bmatrix} +M_{11} & -M_{12} \\ -M_{21} & +M_{22} \end{bmatrix}^T = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Since A is non-singular, $|A| = ad - bc \neq 0$.

As $A^{-1} = \frac{1}{|A|}\text{adj } A$, we get $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Example 1.3

Find the inverse of the matrix $\begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$.

Solution

Let $A = \begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$. Then $|A| = \begin{vmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{vmatrix} = 2(7) + (-12) + 3(-1) = -1 \neq 0$.

Therefore, A^{-1} exists. Now, we get

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